MIL-I-60514A(MU)
20 February 1969
SUPERSEDING
MIL-I-60514(MU)
26 May 1966

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED: DELAY, M49A1 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated: Delay, M49A1, Assembly, loaded with Cartridge, Delay, M90, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167 - Ammunition Data Card
MS-28741 - Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command
F10520232 - Initiator, Cartridge Actuated, Delay, M49A1, Assembly
C8796826 - Container, Ammunition, Fiber, M375, for Initiator Delay, M4A1, M6A1, M12A1, M26, M33, M44, M45A1 or M49A1.

FSC 1377

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DRAWINGS (Cont'd)

C8796827 - Box, Packing, Ammunition, for Initiator, Delay, M4A1, M6A1, M12A1, M26, M43, M44, M45A1 or M49A1, in Fiber Container, M375.
IEL-10520232 - Inspection Equipment List for Initiator, Cartridge Actuated, Delay, M49A1 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

## 3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, size no. 4, as specified in MS-28741.



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TABLE I

| Temperature | Pressure (psi) | Ignition Delay (see 6.3) |
|---|---|---|
| -65° to -70°F | 1300 to 2200 | 0.450 sec., max. |
| 65° to 75°F | 1300 to 2200 | 0.300 ± 0.050 sec. |
| 160° to 165°F | 1300 to 2200 | 0.150 sec., min. |

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.



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4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

| Lot Size | Sample Size | Major | | Minor | |
|---|---|---|---|---|---|
| | | Accept | Reject | Accept | Reject |
| 1 - 300 | 100% Examination | - | - | - | - |
| 301 - 800 | 50 | 0 | 1 | 1 | 2 |
| 801 - 1,800 | 90 | 1 | 2 | 2 | 3 |

4.2.2.1 Cartridge, Delay, M90, Dwg. D8596123, covering a detail of Dwg. F10520232.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Mixed cartridges | Visual |
| 2. Metal defective | Visual |
| 3. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Case scratched, dented, wrinkled or bulged | Visual |
| 102. Varnish missing or inadequate | Visual |



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4.2.2.1 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MINOR** | |
| None defined | |
| 4.2.2.2 Assembly, Dwg. F10520232, less Dwg. D8593314, D8593313 and D8596123. | |
| **CRITICAL** | |
| 1. Mixed initiators | Visual |
| 2. Initiator not in cocked position | Visual |
| 3. Safety pin missing or improperly engaged | Visual |
| **MAJOR** | |
| 101. O-ring missing or not lubricated | Visual |
| 102. Thread damaged | Visual |
| 103. Burr | Visual |
| 104. Foreign matter | |
| **MINOR** | |
| None defined | |
| 4.2.2.3 Assembly, Dwg. F10520232, less Dwg. D8593314 and B8359536. | |
| **CRITICAL** | |
| 1. Cartridge loose | Manual & Visual |
| 2. Cartridge retainer loose | Manual & Visual |
| 3. Metering groove of cartridge retainer missing | Visual |
| 4. Elongated hole of cartridge retainer missing | Visual |
| **MAJOR** | |
| 101. Burr | Visual |
| 102. Foreign matter | Visual |
| **MINOR** | |
| None defined | |



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4.2.2.4 Assembly, Dwg. F10520232, less Dwg. B8359536.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Location of cartridge retainer | Gage |
| 2. Foreign matter in gas port | Visual |
| **MAJOR** | |
| 101. Length from cap boss face to gas port face, max. | Gage |
| 102. Protrusion of firing pin, max. | Gage |
| 103. Set screw missing, loose or above flush | Manual & Visual |
| **MINOR** | |
| None defined | |

4.2.2.5 Assembly, Dwg. F10520232.

| | |
|---|---|
| **CRITICAL** | |
| 1. Base of firing pin housing not in complete contact with face of cartridge | Manual |
| 2. Any part missing, except as otherwise classified | Visual |
| 3. Improper assembly | Visual |
| 4. Marking missing, incorrect or illegible | Visual |
| **MAJOR** | |
| 101. Height of cap, max. | Gage |
| 102. Initiator pin not free to rotate | Manual |
| 103. Safety pin loose or deformed | Manual & Visual |
| 104. Set screw not covered with barrier material | Visual |
| **MINOR** | |
| 201. Total length, max. | Gage |
| 202. Location of labels incorrect | Visual |
| 203. Shipping cap missing | Visual |
| 204. Protective finish inadequate or defective | Visual |



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4.2.2.6 Container, Fiber, Dwg. C8796826.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Gap between body and cover, max. | Manual |
| 202. Contents loose | Manual |
| 203. Metal ends loose or distorted | Manual |
| 204. Cut or damaged through all impregnated layers | Visual |
| 205. Marking missing, incorrect or illegible | Visual |
| 206. Tape incomplete or badly wrinkled | Visual |
| 207. Length of tear tab, min. | Visual |
| 208. Color of tape incorrect | Visual |

4.2.2.7 Box, Packing, Dwg. C8796827.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual |
| 202. Hardware missing, loose, broken or improperly engaged | Manual & Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Manual & Visual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of units incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |



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4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- The initiator chamber and cap of each assembled initiator shall be subjected to the torque application. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the torque application. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional tests.- The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics.- Twenty one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. The variable sampling plan of MIL-STD-414, example C-2 or C03, as applicable, for the AQL listed in Table III, shall be used for each sample of seven units. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 or 3.6.3 or failure of a sample to meet the criteria of the variable sampling plan shall be cause for rejection of the lot.

TABLE III

| Temperature | Sample Size | AQL |
|---|---|---|
| -65° to -70°F | 7 | 0.65% |
| 65° to 75°F | 7 | 0.65% |
| 160° to 165°F | 7 | 0.65% |

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520232.



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4.3 Tests methods and procedures.

4.3.1 Firing pin release.- A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the applicable fixture which registers the amount of pull.

4.3.2 Initiator pin security.- A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the applicable fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.- A torque wrench set at the requirement specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joint. The initiator body shall be secured in an air vise or holding fixture. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south)with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.5.1.1.

4.3.5.1.1 Temperature conditioning.- A standard M49 Initiator, loaded with a dummy M90 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Dwgs. C8796826 and C8796827.



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5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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